

Ước lượng runtime lượng tử

Bảng chứng từ QPU lưu trữ và simulator local

Phạm vi khoa học PARTIAL

Không chạy live QPU. Không gộp các clock khác nhau thành một bảng xếp hạng.

Dữ liệu và phạm vi

Nguồn / panel	Quy mô	Circuit hoặc label được dùng
Real QPU · Ma-Li	340 observations	Logical QASM; observed execution labels
Real QPU · Qonductor	4.482 observations	Submitted physical QASM; one-circuit jobs
Real QPU · QPack	3.945 observations	46 workflows; reconstructed inputs, labels archived
Unified QPU ledger	8.767 observations	Join giữ source ID và row identity; không đồng nhất timing boundary
Simulator local	204 members / 191 hashes	22 families; q2-q16; 162 core + 42 frontier

Local RTX 5070 Ti · ~45 GiB RAM / 16 GiB VRAM. QPU service ≠/schedule ≠/simulator wall-clock.

Thiết kế so sánh

Nguyên tắc	Áp dụng	Giới hạn
Rows chung	Graph V3-large và polynomial dùng cùng unified ledger và grouped outer folds	So sánh theo source-local shared rows
Fit	Preprocessing và calibration chỉ fit trên outer-train	Không chọn theo labels của test
Coverage	Unavailable và failures giữ trong denominator được giao	Coverage khác accuracy
Clock	Hiển thị output clock cùng evaluation target clock	Không xếp hạng chéo clock

MAE thấp hơn trên shared rows không tự động chứng minh superiority trên toàn ledger.

Coverage phương pháp

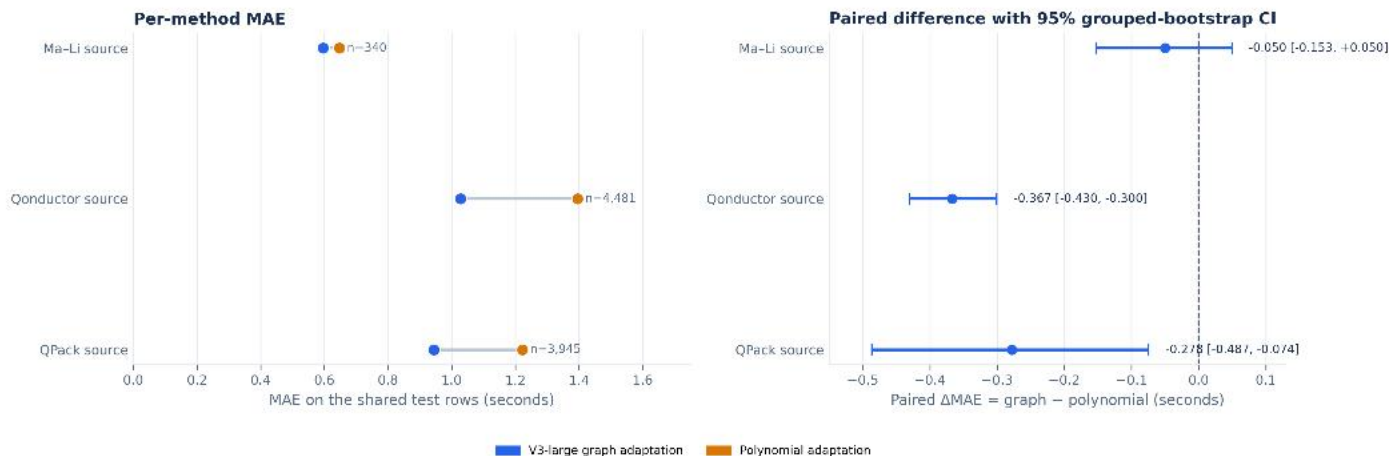
QPU approach	Evidence status	Simulator approach	Evidence status
Ma-Li graph	Unified adaptation; 8,763–8,766 / 8,767	Ma-Li-style graph	Aer + fixed-MPS local adaptations
Qonductor	Unified polynomial adaptation; 8,767 / 8,767	Azizov-style	Source / hybrid / transpiled GNN + baselines
Scholten	Nominal proxy; 5,769 / 8,767	Family-Aware-style	MPS runtime-only residual; joint target unavailable
QCRE	Scheduled proxy; 8,766 / 8,767	Maestro	Pilot gate failed; no predictor score
Qiskit estimate_duration	Native schedule; 8,766 / 8,767	cuTensorNet	Native same-plan estimate / actual diagnostic
Hyb-HANAS	One-circuit cost adaptation; 8,420 finite	Pasqal emu-MPS	Analog companion; not digital common panel

MPS runtime-only adaptation was evaluated; 24 QPU variants are not 24 papers. Coverage $\neq$ accuracy.

MAE theo source trên shared rows

Real-QPU source-local paired error

V3-large graph adaptation vs unified polynomial · lower MAE is better · same successful test IDs per source



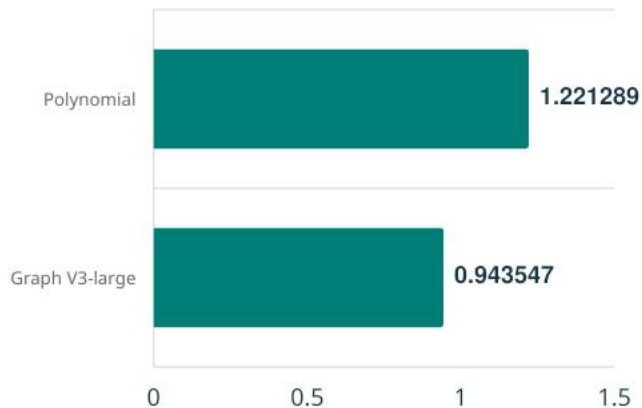
Source slices are descriptive, not separate source-trained models. Ma-Li CI crosses zero; Qconductor coverage is asymmetric (row477 unavailable).

<https://arxiv.org/abs/2404.18015> SHA-256: 4bfeff9637a48b7e8b2a918c0b151701d730ed574b5d8c685c60fab0b1111

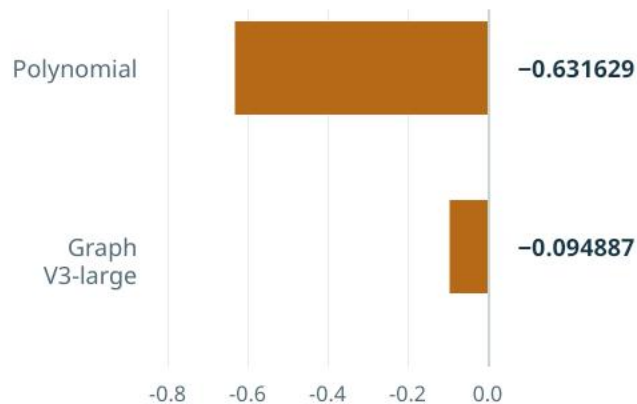
Shared rows by source. 95% grouped-bootstrap CI for graph minus polynomial MAE. Lower MAE favors the graph adaptation.

QPack: MAE và R^2

MAE (s)



R^2



Equal-workflow MAE diagnostic: 0.946290 s vs 1.217754 s across 46 workflows. Both R^2 values remain negative.

6 templates recur; workflow holdout is not unseen-template validation.

Scheduled duration và observed service time

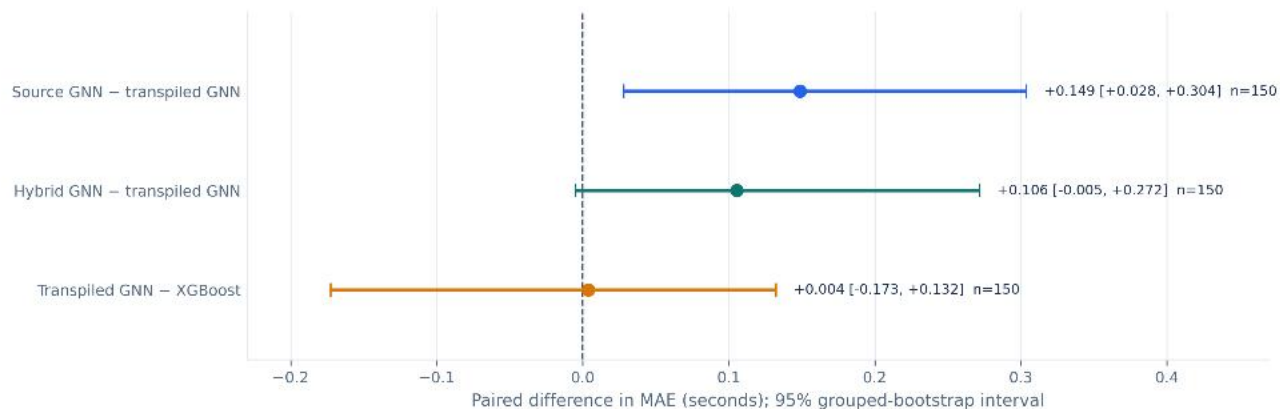
Route	Output clock	What it excludes or changes
QCRE critical path	Scheduled single-shot	No queue, service, transport, or fixed overhead
Qiskit estimate_duration	Scheduled single-shot	Native API output under frozen Target
Shot-scaled variants	Scheduled total proxy	Non-native multiplication by shots
Outer-train calibration	Calibrated service-time target	Adaptation, fit on outer-train only

8.766 predictions per raw route. Row4477 is unavailable. Scheduled duration does not replace observed QPU service time.

Aer: predictor trên 150 circuit hashes

Aer warm runtime: paired error contrasts

One local Aer target · 150/150 shared circuit hashes · ΔMAE = candidate – reference (seconds)



MAE: source 0.450s · hybrid 0.406s · transpiled 0.301s · XGBoost(transpiled) 0.297s. Source-vs-transpiled interval excludes zero; hybrid-vs-transpiled and GNN-vs-XGBoost intervals include zero.

Intervals are unadjusted 95% percentile intervals; they do not establish universal superiority or transfer to other Aer targets.

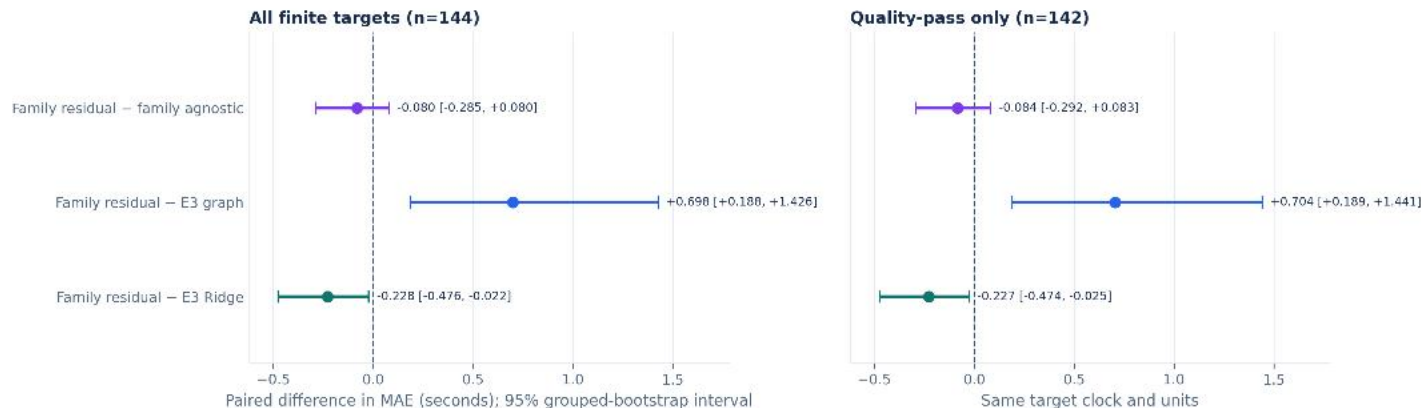
Data: predictive runtime aggregate v1 · source hashes in RFADMC.md

Noisy-Aer warm · n=150. MAE: source 0.450 s, hybrid 0.406 s, transpiled 0.301 s, XGBoost 0.297 s; GNN-XGBoost Δ unresolved.

CUDA-Q MPS: fixed-configuration runtime predictor

Fixed- χ 16 CUDA-Q MPS: runtime-model contrasts

FP64, bond dimension 16 • evaluation target is MPS warm state execution • Δ MAE = candidate – reference (seconds)



All-finite MAE: E3 graph 0.302s • E5 family residual 1.001s • E5 family agnostic 1.081s • E3 Ridge 1.229s. Family residual vs family-agnostic interval crosses zero; family-residual vs E3 graph is positive (worse).

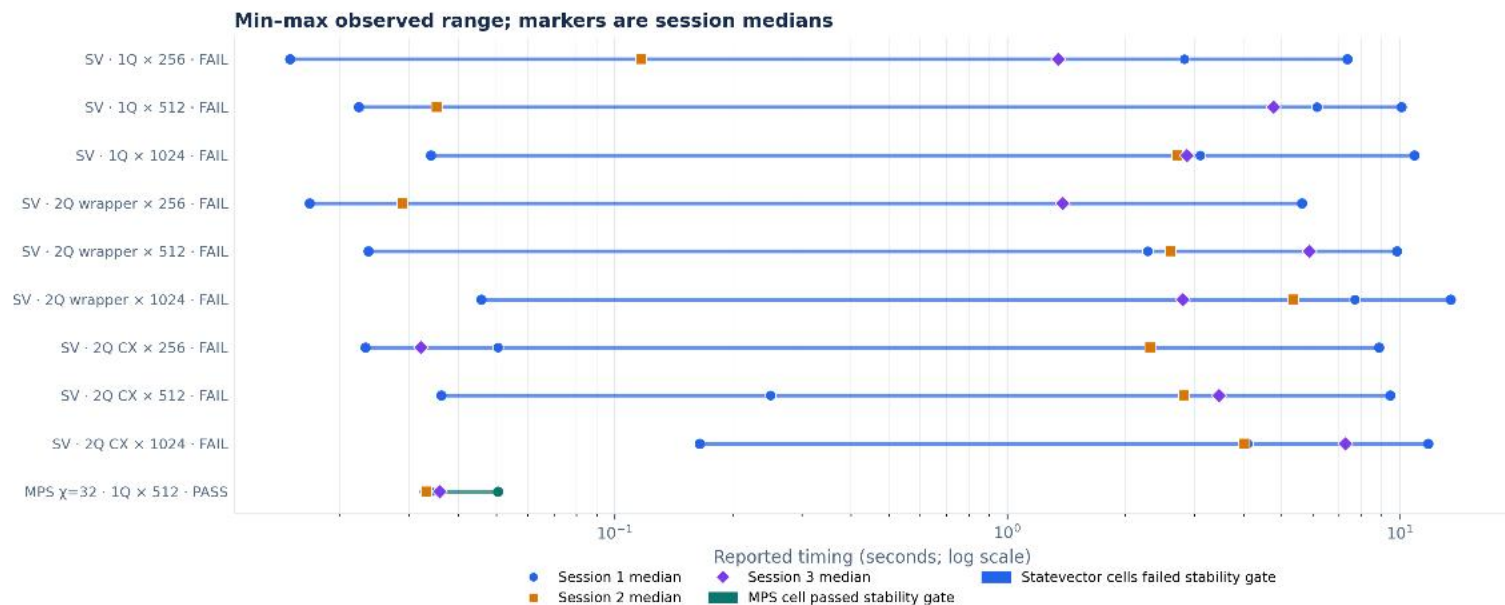
The MPS quality gate is fidelity ≥ 0.99 . All-finite includes two finite but quality-failed circuits; quality-pass is a prespecified subset. Intervals are unadjusted.

Data: predictive-runtime-aggregate-v1 • source hashes in README.md

MPS FP64 • bond 16 • warm state • 144 finite / 142 quality-pass. Graph MAE 0.304 s; max error 25.10 s. Family benefit unresolved.

Maestro pilot timing spread

10 synthetic calibration cells · process-isolated reported execution · timing stability gate failed overall



Terminal pilot: 9/9 statevector cells failed stability; 1/1 MPS cell passed. No accepted predictor or panel accuracy score; cause of spread is undetermined.

simulator/maestro_pilot_summary.csv 5f1A 256 d6107c87721695537cee9c67f3e7c51ab916f620a665b532487d1374e7d02ad5

Kết luận và phạm vi còn thiếu

Có thể kết luận	Chưa đủ evidence để kết luận
Graph V3-large có MAE thấp hơn polynomial trên source-local shared rows.	Không có pooled cross-source superiority claim.
QPack có MAE thấp hơn nhưng cả hai R^2 đều âm.	Không nói model giải thích tốt biến thiên runtime.
E6 có Aer GNN và fixed-MPS predictor OOF results.	Chỉ một Aer config; MPS một bond/config; CI exploratory.
Maestro pilot ghi nhận stability failure.	Không có accepted Maestro predictor hoặc panel score.

Scientific coverage: PARTIAL. Build validation không đồng nghĩa full replication hoặc public-release approval.

Fidelity phương pháp và tái dựng đầu vào

Mảng	Đã dùng	Giới hạn fidelity
QPU graph	Ma-Li/QPack logical; Qonductor exact submitted physical QASM	Không thực thi compiled-input promise cho Ma-Li/QPack
Qonductor recovery	230 archive-resolved logical recipes	Recipe resolution, not byte-exact original logical QASM
QPack replay	6 reconstructed structures; representative rz(0.3)/rx(0.2)	Không có original optimizer angles / submitted routing
Graph features	7 global features; DAG + global MLP branches	Polynomial có 5 features; không cô lập causal graph effect
Native estimates	Snapshot calibration theo backend	Không phải row-day historical calibration
Family-Aware-style	Fixed-MPS runtime-only residual + matched ablation	Original joint threshold/runtime target not evaluated
Maestro	Component calibration / local context pilot	No accepted held-out predictor score; pilot gate failed

Archived labels vẫn là runtime quan sát; replay reconstruction không thay đổi labels.

Clocks, quality, và release boundary

Domain	Clock or outcome	Interpretation
QPU target	Archived observed service/execution	Evaluation target, not schedule duration
QPU analytical	Scheduled single-shot / shot-scaled	Proxy; provider overhead excluded
Simulator	First / warm execution; quality gate	MPS: 573/612 ok; 21 quality fail, 18 adapter error
cuTensorNet	Selected-plan estimate	603/612 ok; compare only to same-plan contraction
Maestro	Process-isolated reported execution	Fresh process, terminal pilot failure
Reproducibility	Local clean clone at 461cc6e	Tables/E6 rebuilt; current reporting edits need final C6 check
Rights/provenance	Owner decision remains open	Not a public-release approval; see third-party notices

Unavailable rows, timeouts, adapter errors, and quality failures stay in coverage denominators.

Appendix: coverage của 24 QPU variants

Variant	Pred / 8,767	Cover	Output clock
MaLi graph V3	8,763	99.95%	Observed service
MaLi graph V3-large	8,766	99.99%	Observed service
Unified polynomial V3	8,767	100.00%	Observed service
QCRE CP raw	8,766	99.99%	Scheduled CP
QCRE CP affine	8,766	99.99%	Cal. service
QCRE CP log-affine	8,766	99.99%	Cal. service
QCRE shot raw	8,766	99.99%	Shot schedule
QCRE shot affine	8,766	99.99%	Cal. service
QCRE shot log-affine	8,766	99.99%	Cal. service
Qiskit duration raw	8,766	99.99%	Scheduled 1-shot
Qiskit duration affine	8,766	99.99%	Cal. service
Qiskit duration log-affine	8,766	99.99%	Cal. service

Variant	Pred / 8,767	Cover	Output clock
Qiskit shot raw	8,766	99.99%	Shot schedule
Qiskit shot affine	8,766	99.99%	Cal. service
Qiskit shot log-affine	8,766	99.99%	Cal. service
Hyb 1-circuit raw	8,420	96.04%	T_eff
Hyb 1-circuit affine	8,420	96.04%	Cal. service
Hyb 1-circuit log-affine	8,420	96.04%	Cal. service
Hyb shot-linear raw	8,420	96.04%	Shot-linear cost
Hyb shot-linear affine	8,420	96.04%	Cal. service
Hyb shot-linear log-affine	8,420	96.04%	Cal. service
Scholten nominal raw	5,769	65.80%	CLOPSv-like
Scholten affine	5,769	65.80%	Cal. service
Scholten log-affine	5,769	65.80%	Cal. service